

This assignment will not be graded and is only for practice.

Level 1

1) Choose the set of correct options.

1 point

- ☐ If S is a spanning set of the vector space V , then $S \cup \{v\}$ must be a spanning set of V , for all $v \in V$.
- ☐ Span of an empty set is the zero vector space.
- ☐ If S is a spanning set of the vector space V , then $S \setminus \{v\}$ must be a spanning set of V , for all $v \in S$.
- ☐ If S is a spanning set of the vector space V , then $S \cup \{v\}$ may not be a spanning set of V , for all $v \in V$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If S is a spanning set of the vector space V , then $S \cup \{v\}$ must be a spanning set of V , for all $v \in V$.

Span of an empty set is the zero vector space.

2) Let S be the set of matrices where $S = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$. Consider the following statements:

Statement P: $\text{Span}(S)$ is the vector space consisting of only lower triangular square matrices of order 2.

Statement Q: $\text{Span}(S)$ is the vector space consisting of only upper triangular square matrices of order 2.

Statement R: $\text{Span}(S)$ is the vector space consisting of all the square matrices of order 2.

Statement S: $\text{Span}(S)$ is the vector space consisting of only scalar matrices of order 2.

Find the number of correct statements.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

3) If S is a basis of $\mathbb{R}^2$, then what is the cardinality of S ?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

4) Let $S = \{(1, 2), (1, 3)\}$ and $\text{span}(S) = \mathbb{R}^n$. Then find the value of n .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

5) Let S be the set of matrices where $S = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$. Consider the following statements:

Statements P: $\text{Span}(S)$ is the vector space consisting of only the Identity matrix of order 2.

Statements Q: $\text{Span}(S)$ is the vector space consisting of only diagonal matrices of order 2.

Statements R: $\text{Span}(S)$ is the vector space consisting of all the square matrices of order 2.

Statements S: $\text{Span}(S)$ is the vector space consisting of only scalar matrices of order 2.

Find the number of correct statements.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

6)

1 point

Consider the following set of vectors $S =$

$\{(1, 1, -1), (-1, 1, 1), (0, \frac{1}{2}, 0), (0, 1, -2), (1, 0, -2)\}$. Choose the set of correct options.

- ☐ The set $\{(1, 1, -1), (-1, 1, 1), (0, \frac{1}{2}, 0)\}$ is not a basis of $\mathbb{R}^3$.
- ☐ The set $\{(0, \frac{1}{2}, 0), (0, 1, -2), (1, 0, -2), (1, 0, -2)\}$ is a basis of $\mathbb{R}^3$.
- ☐ The set $\{(-1, 1, 1), (0, \frac{1}{2}, 0), (0, 1, -2)\}$ is a basis of $\mathbb{R}^3$.
- ☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The set $\{(1, 1, -1), (-1, 1, 1), (0, \frac{1}{2}, 0)\}$ is not a basis of $\mathbb{R}^3$.

The set $\{(-1, 1, 1), (0, \frac{1}{2}, 0), (0, 1, -2)\}$ is a basis of $\mathbb{R}^3$.

7) Consider a set of vectors $S = \{(1, 2), (-1, 2), (3, 1), (-3, 1)\}$ in $\mathbb{R}^2$. Choose the set **1 point** of correct options.

- ☐ The set $\{(1, 2), (-1, 2), (3, 1)\}$ is a basis of $\mathbb{R}^2$.
- ☐ The set $\{(-1, 2), (3, 1)\}$ is a basis of $\mathbb{R}^2$.
- ☐ The set $\{(3, 1), (-3, 1)\}$ is a basis of $\mathbb{R}^2$.
- ☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The set $\{(-1, 2), (3, 1)\}$ is a basis of $\mathbb{R}^2$.

The set $\{(3, 1), (-3, 1)\}$ is a basis of $\mathbb{R}^2$.

Level 2

8) Choose the set of correct options.

1 point

- ☐ If a subset S of $\mathbb{R}^2$ contains only two elements then S is a basis of $\mathbb{R}^2$.
- ☐ If a subset S of $\mathbb{R}^3$ contains only one elements then S can never be a basis of $\mathbb{R}^3$.
- ☐ If a subset S of $\mathbb{R}$ contains only one non zero element then S is a basis of $\mathbb{R}$.
- ☐ If S_1 and S_2 are two bases of $\mathbb{R}^3$ then $S_1 = S_2$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If a subset S of $\mathbb{R}^3$ contains only one elements then S can never be a basis of $\mathbb{R}^3$.

If a subset S of $\mathbb{R}$ contains only one non zero element then S is a basis of $\mathbb{R}$.

9) Let V be the subspace of $\mathbb{R}^3$ defined as follows:

1 point

$$V = \{(x, y, z) \mid x = y - z, \text{ and } x, y, z \in \mathbb{R}\}$$

Choose the set of correct options from the following.

- ☐ $\{(1, 1, 0), (1, 0, -1)\}$ is a linearly independent set of V .
- ☐ $\{(1, 1, 0), (1, 0, -1), (0, 1, 1)\}$ is a linearly independent set of V .
- ☐ $\{(0, 1, 1), (1, 0, -1)\}$ is a spanning set of V .
- ☐ $\{(1, 1, 0)\}$ is a spanning set of V .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(1, 1, 0), (1, 0, -1)\}$ is a linearly independent set of V .

$\{(0, 1, 1), (1, 0, -1)\}$ is a spanning set of V .

10) Let V and W be vector spaces which are defined as follows:

1 point

$V = \{(x, y) \mid y = mx, \text{ where } m = 0 \text{ and } x, y, m \in \mathbb{R}\}$ with usual addition and scalar multiplication as in $\mathbb{R}^2$. $W = \{(x, y) \mid x = 0\}$ with usual addition and scalar multiplication as in $\mathbb{R}^2$.

Choose the correct set of options.

- ☐ The set $\{(1, m), (\frac{1}{m}, 1)\}$ is a linearly independent set in V .
- ☐ The set $\{(1, m), (\frac{1}{m}, 1)\}$ is a spanning set for V .
- ☐ The set $\{(1, m)\}$ is a linearly independent set in V .
- ☐ The set $\{(\frac{1}{m}, 1)\}$ is a linearly independent set in V .
- ☐ The set $\{(0, 1), (0, 2)\}$ is a linearly independent set in W .
- ☐ The set $\{(0, 1)\}$ is a linearly independent set in W .
- ☐ The set $\{(0, 5)\}$ is a spanning set for W .

No, the answer is incorrect.

Score: 0

Accepted Answers:

The set $\{(1, m), (\frac{1}{m}, 1)\}$ is a spanning set for V .
 The set $\{(1, m)\}$ is a linearly independent set in V .
 The set $\{(\frac{1}{m}, 1)\}$ is a linearly independent set in V .
 The set $\{(0, 1)\}$ is a linearly independent set in W .
 The set $\{(0, 5)\}$ is a spanning set for W .

Your score is: 0/10